In [18]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_diabetes
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.linear_model import Ridge, Lasso, ElasticNet
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

data = load_diabetes()
X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target, name="target")

print("Dataset shape:", X.shape)
print(X.head())

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])

ridge_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", Ridge())
])

lasso_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", Lasso(max_iter=10000))
])

elastic_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", ElasticNet(max_iter=10000))
])

ridge_params = {
    "model__alpha": [0.001, 0.01, 0.1, 1, 10, 100]
}

ridge_grid = GridSearchCV(
    ridge_pipeline, ridge_params, cv=5,
    scoring="neg_mean_squared_error"
)

ridge_grid.fit(X_train, y_train)

print("Best Ridge parameters:", ridge_grid.best_params_)

lasso_params = {
    "model__alpha": [0.001, 0.01, 0.1, 1, 10]
```

```
}

lasso_grid = GridSearchCV(
    lasso_pipeline, lasso_params, cv=5,
    scoring="neg_mean_squared_error"
)

lasso_grid.fit(X_train, y_train)

print("Best Lasso parameters:", lasso_grid.best_params_)

elastic_params = {
    "model__alpha": [0.001, 0.01, 0.1, 1, 10],
    "model__l1_ratio": [0.1, 0.3, 0.5, 0.7, 0.9]
}

elastic_grid = GridSearchCV(
    elastic_pipeline, elastic_params, cv=5,
    scoring="neg_mean_squared_error"
)

elastic_grid.fit(X_train, y_train)

print("Best Elastic Net parameters:",
      elastic_grid.best_params_)

ridge_pred = ridge_grid.predict(X_test)
lasso_pred = lasso_grid.predict(X_test)
elastic_pred = elastic_grid.predict(X_test)

results = pd.DataFrame({
    "Model": ["Ridge", "Lasso", "Elastic Net"],
    "MSE": [
        mean_squared_error(y_test, ridge_pred),
        mean_squared_error(y_test, lasso_pred),
        mean_squared_error(y_test, elastic_pred)
    ],
    "MAE": [
        mean_absolute_error(y_test, ridge_pred),
        mean_absolute_error(y_test, lasso_pred),
        mean_absolute_error(y_test, elastic_pred)
    ],
    "R2": [
        r2_score(y_test, ridge_pred),
        r2_score(y_test, lasso_pred),
        r2_score(y_test, elastic_pred)
    ]
})

print(results)

lasso_model = lasso_grid.best_estimator_.named_steps["model"]

lasso_coefficients = pd.Series(
    lasso_model.coef_, index=X.columns
)
```

```
print(lasso_coefficients)
print("Selected features:")
print(lasso_coefficients[lasso_coefficients != 0])
```

```
Dataset shape: (442, 10)
        age       sex       bmi        bp        s1        s2        s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163  0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596  0.008142

         s4        s5        s6
0 -0.002592  0.019907 -0.017646
1 -0.039493 -0.068332 -0.092204
2 -0.002592  0.002861 -0.025930
3  0.034309  0.022688 -0.009362
4 -0.002592 -0.031988 -0.046641
Training samples: 353
Testing samples: 89
Best Ridge parameters: {'model__alpha': 10}
Best Lasso parameters: {'model__alpha': 1}
Best Elastic Net parameters: {'model__alpha': 0.1, 'model__l1_ratio': 0.1}
         Model          MSE        MAE        R2
0        Ridge  2875.778718  42.856825  0.457211
1        Lasso  2824.568094  42.802984  0.466877
2  Elastic Net  2860.948451  42.949530  0.460010
age     0.687032
sex    -9.297519
bmi    26.219225
bp     15.657314
s1     -8.228172
s2     -0.000000
s3     -9.024087
s4      3.420861
s5     22.636465
s6      2.098647
dtype: float64
Selected features:
age     0.687032
sex    -9.297519
bmi    26.219225
bp     15.657314
s1     -8.228172
s3     -9.024087
s4      3.420861
s5     22.636465
s6      2.098647
dtype: float64
```

In [16]:
```
!jupyter nbconvert --to html /content/smld_6_new.ipynb
```

```
[NbConvertApp] Converting notebook /content/smld_6_new.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 2 image(s).
[NbConvertApp] Writing 346999 bytes to /content/smld_6_new.html
```